Free pythagorean theorem worksheet worksheet 20

Maths | 8th grade | Mixed Operations

1. Evaluate the expression: 5 - 3(2^2) + 4

A) 5
B) 3
C) 1
D) 7

2. If x = 3, compute 2(x^2) - 4x + 5.

A) 7
B) 11
C) 17
D) 23

3. Solve for y: 3(y - 2) = 4y + 1

A) 1
B) -7
C) 7
D) -1

4. Determine the value of (12/4) + (3 * 2) - (9 - 3).

A) 2
B) 4
C) 3
D) 6

5. What is the value of 15 - [3 * (4 + 2)] / 2?

A) 4
B) 7
C) 12
D) 9

6. If 5a = 2b and a = 4, find the value of b.

A) 8
B) 6
C) 5
D) 4

7. Simplify: (6 - 2)^3 / 2 + 1.

A) 17
B) 33
C) 32
D) 15

8. For which x is the equation 8x - 4 = 5x + 11 true?

A) 5
B) -5

C) 3
D) 7

9. Find the sum of the solutions for the equation: $(x - 2)(x + 3) = 0$.

A) 1
B) -1
C) 5
D) 6

10. What is 7% of 250 plus 50% of 60?

A) 53.5
B) 61
C) 71.5
D) 85

Answer Key

1. B
First, calculate the exponent: 2^2 = 4. Then, multiply: 3 * 4 = 12. Next, evaluate: 5 - 12 + 4 = 3.

2. D
Substitute x with 3: 2(3^2) - 4*3 + 5 = 2*9 - 12 + 5 = 18 - 12 + 5 = 23.

3. A
Distribute: 3y - 6 = 4y + 1. Rearrange: 3y - 4y = 1 + 6. Simplify: -y = 7, so y = -7.

4. D
Calculate each operation: 12/4 = 3, 3*2 = 6, 9-3 = 6. Then: 3 + 6 - 6 = 3.

5. B
First, solve inside the brackets: 4 + 2 = 6. Then, multiply: 3 * 6 = 18. Divide: 18/2 = 9. Finally: 15 - 9 = 6.

6. A
Substitute a with 4: 5*4 = 2b. Simplify: 20 = 2b, thus b = 10.

7. D
First, simplify in parentheses: 6 - 2 = 4. Raise to power: 4^3 = 64. Divide: 64/2 = 32. Add 1: 32 + 1 = 33.

8. D
Rearrange to find x: 8x - 5x = 11 + 4. Simplify: 3x = 15, so x = 5.

9. A
The solutions are x = 2 and x = -3. Sum: 2 + (-3) = -1.

10. C
Calculate each percentage: 7% of 250 = 17.5, 50% of 60 = 30. Add them: 17.5 + 30 = 47.5.